



Semester I Examinations 2018/19

Exam Code(s)	1SD1, 1SD3, 1MF1
Exam(s)	Higher Diploma (Software Design & Development) M.Sc. (Software Design & Development) Higher Diploma (Software Design & Development) – Industry Stream
Module Code(s)	CT516 CT861 CT538 CT542
Module(s)	Computer Architecture and Operating Systems
Paper No.	1
Repeat Paper	No
External Examiner(s)	Professor Pier Luca Lanzi
Internal Examiner(s)	Prof. Michael Madden *Dr. Michael Schukat

Instructions: Answer **question 1** and any other question from Section A, and one question from Section B.

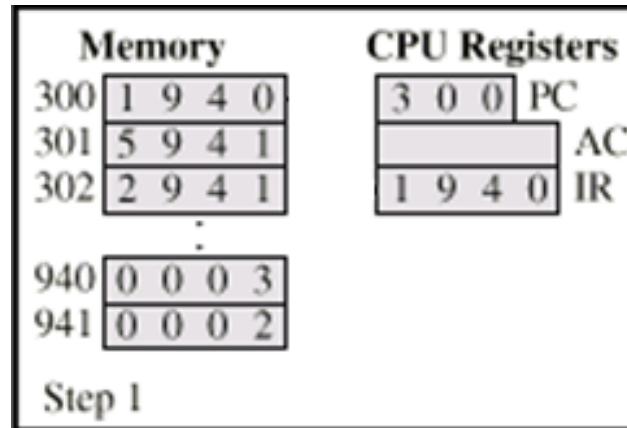
Duration	2 hrs
No. of Pages	5
Discipline(s)	Information Technology
Requirements	None

Release to Library: Yes ☒ No ☐

Section A

Question 1 (COMPULSARY)

- a) What is meant by the *von Neuman bottleneck*? Using diagrams show how the Harvard architecture or the use of caches alleviates this problem. 3 MARKS
- b) Consider the following example of program execution discussed in the lectures;



which is based on the following instruction set:

- $1xxx_{16}$: load AC from memory location xxx
- $2xxx_{16}$: store AC to memory location xxx
- $5xxx_{16}$: add to AC from memory location xxx

Extend the instruction set and write a program that calculates the expression

$$(7 * (3 + 4)) - 10$$

and stores the result at memory address 950_{16} . Fully explain your solution.

5 MARKS

- c) Using an example outline in detail how the *paged memory allocation* scheme works. In your answer refer to the terms, *pages*, *frames*, *page tables*, and *demand paging*. 5 MARKS
- d) Show how an "if-then-else" block in a high-level programming language such as Java may be translated into a series of CPU instructions. In your answer explain the concept of the *conditional jump* instruction and the *unconditional jump* instruction. 4 MARKS
- e) Distinguish between the following addressing modes:
- Immediate Addressing
 - Direct Addressing
 - Indirect Addressing
- 3 MARKS

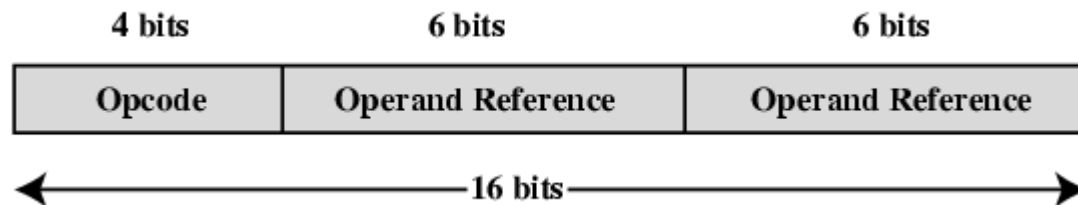
PTO

Question 2

- a) What is a *multi-core microprocessor*? Use a diagram to show the typical arrangement of (level-1 / level-2) caches in such an architecture.

3 MARKS

- b) The diagram below shows a hypothetical instruction format:



- How many *opcodes* and *operand references* does this design support?
- Revise the layout so that it supports 136 opcodes, three operand references and a total instruction length of 32 bits.

4 MARKS

- c) Distinguish between *interrupt-driven* and *DMA-driven* data input/output. Why would you choose one over the other?

4 MARKS

- d) What is a *stack* and what operations can be performed on a stack? Use examples to illustrate your answer.

3 MARKS

- e) Explain why *cache memory* is based on static rather than dynamic RAM.

2 MARKS

- f) Summarise the *7-state process model* and, using an example, show how a process moves through all different states.

4 MARKS

PTO

Question 3

- a) What is *Moore's Law* and how did it impact on microprocessor development over the last 40 years?

3 MARKS

- b) Distinguish between RISC and CISC and show (using hypothetical instruction sets) how a CISC instruction can be translated into a series of RISC instructions.

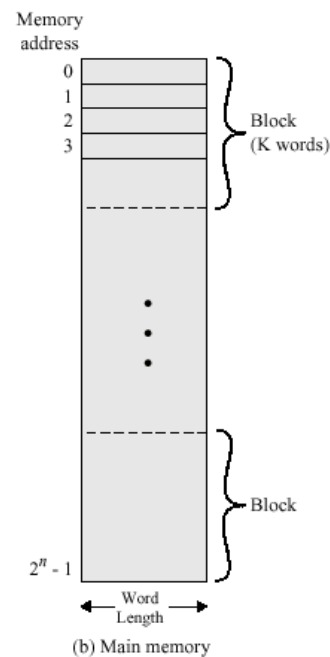
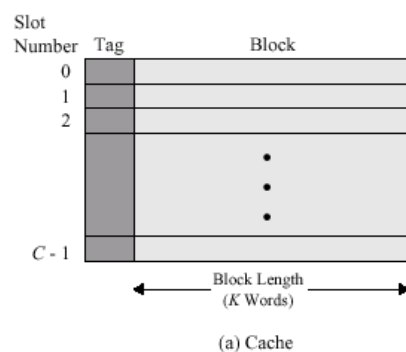
2 MARKS

- c) Using code snippets and / or diagrams show how two concurrent processes can use a semaphore to control their access to a non-sharable or critical resource.

5 MARKS

- d) Explain the two diagrams below and provide a detailed step-by-step example on the operation of a cache.

Further on, distinguish between the *write-back* and *write-through* caching strategies.



5 MARKS

- e) Explain the inner workings of a hard disk with multiple platters. In your answer refer to the terms *sector*, *track* and *cylinder*.

2 MARKS

- f) Using examples, distinguish between the following file allocation methods:
 - indexed allocation
 - chained allocation

3 MARKS

PTO

Section B

Question 4

- a) What is meant by the *client-server model*? Explain the *message request / reply mechanism* used in this model.
3 MARKS
- b) Summarise the inner workings and disadvantages / limitations of the ALOHA multiple access protocol.
5 MARKS
- c) Discuss the purpose and inner workings of the following protocols:
a. Dynamic Host Configuration Protocol (DHCP)
b. Address Resolution Protocol (ARP)
6 MARKS
- d) What is the purpose of the *transport layer* in a network protocol stack? Briefly summarise the features and header structure of the *UDP transport layer protocol*.
6 MARKS

Question 5

- a) Using a diagram summarise the typical structure of a 802.3 MAC sublayer protocol frame. What kind of a source / destination addressing scheme would be used in this layer?
6 MARKS
- b) Briefly outline the similarities and differences between a *hub* and a *switch*.
2 MARKS
- c) Using a simple example summarise the inner workings of the shortest path routing algorithm.
7 MARKS
- d) What is the purpose of subnets and subnet masking in an IP network. Use examples to illustrate your answer.
5 MARKS